



Department of Information Technology

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech Information Technology

Course Code & Title: CD3291 Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari, AP /IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Recursion
- **Date of Implementation:** 25.09.2023
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Think-Pair-Share
- **Justification:**
 - Recursion is defined as function that calls itself. In the program, same function will be available inside the main() and to the function itself. To test the student's understanding of recursion programming, think pair share activity is chosen. This practice assists students in gaining a thorough understanding of this topic. Students think about various concepts that work based on recursion and express their thoughts with their classmates, which helps them enhance their creative thinking and oral communication abilities.

Time Allotted for the Activity:

- **Procedure of the Implementation:**
 - Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).
 - T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-5 Minutes].
 - P (Pair): Teacher asks another question, related to the previous one, that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 2-5 Minutes].
 - S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 05-10 minutes].

Details of the Implementation

- First of all, teacher taught the topics and provide the materials to think and discuss.
- Teacher provide a different application based questions about the topic and ask them to form a group and gave a 5 minutes' time to think. The Students come up with their own individual answer to the question.



- Finally, the students shared their ideas and answers based on their application based questions.
- The questions discussed based on the recursion concept was Fibonacci, Tower of Hanoi and English Ruler.
- The Figure 1 shows the Think Pair Share activity done by the students.
- Finally, teacher and students discussed the answers of all questions at the end of the activity that is shown in Figure 2.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO9	PSO10
CO1	3	3	3	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative Practice	PO1	PO2	PO3	PO5	PO9	PSO1
Justification for correlation	This activity helps the students to gain the programming knowledge on recursion	Students will be able to analyse the problem and determine when to apply recursion function	Students will be able to provide solution by using recursive functions	Students were asked to use online python compiler for executing the concept	The students coordinate and demonstrate the activity as a team.	The recollecting and discussing skill earned through this activity helps the students in solving problems related to recursion

- **Images / Screenshot of the practice:**



Figure 1 – Students Think and Share their Thoughts with team mates



Figure2– Student Abinaya. S. G shared about her thoughts about recursion

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- ❖ Most of the students actively participated and enjoyed. Students were given a solution with various formats.
- ❖ Most of the students students enjoyed with peer learning.
- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - Students are actively engaged.
 - Students learn from each other.
 - Makes class interactive.
 - Builds a friendly, yet academic atmosphere.
 - Includes all the students in the teaching-learning process.

❖ ***Challenges faced in implementation:***

- Some students were struggled to participate in TPS.
- Noise level of the class room goes up and a transition was needed to get backon the topic.
- It was hard to keep every student on task.
- It took a longer time than other strategies.

References:

- <https://www.readingrockets.org/strategies/think-pair-share>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1.TPS_AI_GM.pdf
- [https://www.ritrjpm.ac.in/images/computer-science/2021 2022/ MGM VI Think Pair and_Share.pdf](https://www.ritrjpm.ac.in/images/computer-science/2021%202022/MGM%20VI%20Think%20Pair%20and%20Share.pdf)
- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/BV_CS8602_Think-Pair-Share.pdf